MAT 4233 — Weekly Schedule

Modern Abstract Algebra · Fall 2026 · UT San Antonio

Texts

Primary text

- **KvP** — Kramer & von Pippich, *From Natural Numbers to Quaternions*
- **Beardon** — Beardon, *Algebra and Geometry*

Supplementary reading

- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*

Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*

Schedule

Date	Topic	Reading
Thu 20 Aug	Lecture — Course launch; divisibility and the division algorithm	KvP §1.1-1.2
Tue 25 Aug	Lecture — Primes, the Fundamental Theorem of Arithmetic, gcd and lcm	KvP §1.3-1.5
Thu 27 Aug	Lecture — Semigroups, monoids, groups — all of them Z/nZ	KvP §2.1
Tue 1 Sep	Lecture — Subgroups; the unit group (Z/nZ)* (Due: Set 1)	KvP §2.2
Thu 3 Sep	**Q&A** — Defense 1	
Tue 8 Sep	Lecture — Group homomorphisms; the order of an element	KvP §2.3
Thu 10 Sep	Lecture — Cyclic groups; primitive roots	KvP §2.3
Tue 15 Sep	Lecture — Cosets and Lagrange's theorem (Due: Set 2)	KvP §2.4
Thu 17 Sep	**Q&A** — Defense 2	
Tue 22 Sep	Lecture — Normal subgroups and quotient groups	KvP §2.5
Thu 24 Sep	Lecture — The homomorphism theorem	KvP §2.6
Tue 29 Sep	Lecture — Euler's theorem, fast modular exponentiation, and RSA end to end	KvP App. B; KvP §2.4
Thu 1 Oct	**Exam** — Midterm Exam 1 — Segment 1	
Tue 6 Oct	Lecture — Rings and subrings	KvP §3.2
Thu 8 Oct	Lecture — Ring homomorphisms	KvP §3.3
Thu 15 Oct	Lecture — Ideals	KvP §3.3
Tue 20 Oct	Lecture — Quotient rings (Due: Set 3)	KvP §3.3

Date	Topic	Reading
Thu 22 Oct	**Q&A** — Defense 3	
Tue 27 Oct	Lecture — Fields and skew fields; F_p	KvP §3.4
Thu 29 Oct	Lecture — Integral domains and the field of fractions	KvP §3.5
Tue 3 Nov	Lecture — Unique factorization: UFDs (Due: Set 4)	KvP §3.7
Thu 5 Nov	**Q&A** — Defense 4	
Tue 10 Nov	Lecture — PIDs and Euclidean domains	KvP §3.7
Thu 12 Nov	Lecture — Complex numbers: arithmetic, the plane, polar form, roots of unity	Beardon §3.1-3.2, §3.5; KvP §5.1-5.2
Tue 17 Nov	Lecture — Mobius transformations: the formula, composition as matrix multiplication, circles to circles, the Riemann sphere, fixed points (Due: Set 5)	Beardon Ch. 13; Arnold–Rogness; Needham Ch. 3 §I-II, §IV-VI (handout; a copy is on reserve); Mobius Transformations Revealed (2 min)
Thu 19 Nov	**Q&A** — Defense 5	
Tue 24 Nov	Lecture — The Gaussian integers: norm, division with remainder, irreducibles, and sums of two squares — a survey	Conrad
Tue 1 Dec	*Review* — Review — the exam pool	
Thu 3 Dec	**Exam** — Midterm Exam 2 — Segment 2 weighted	